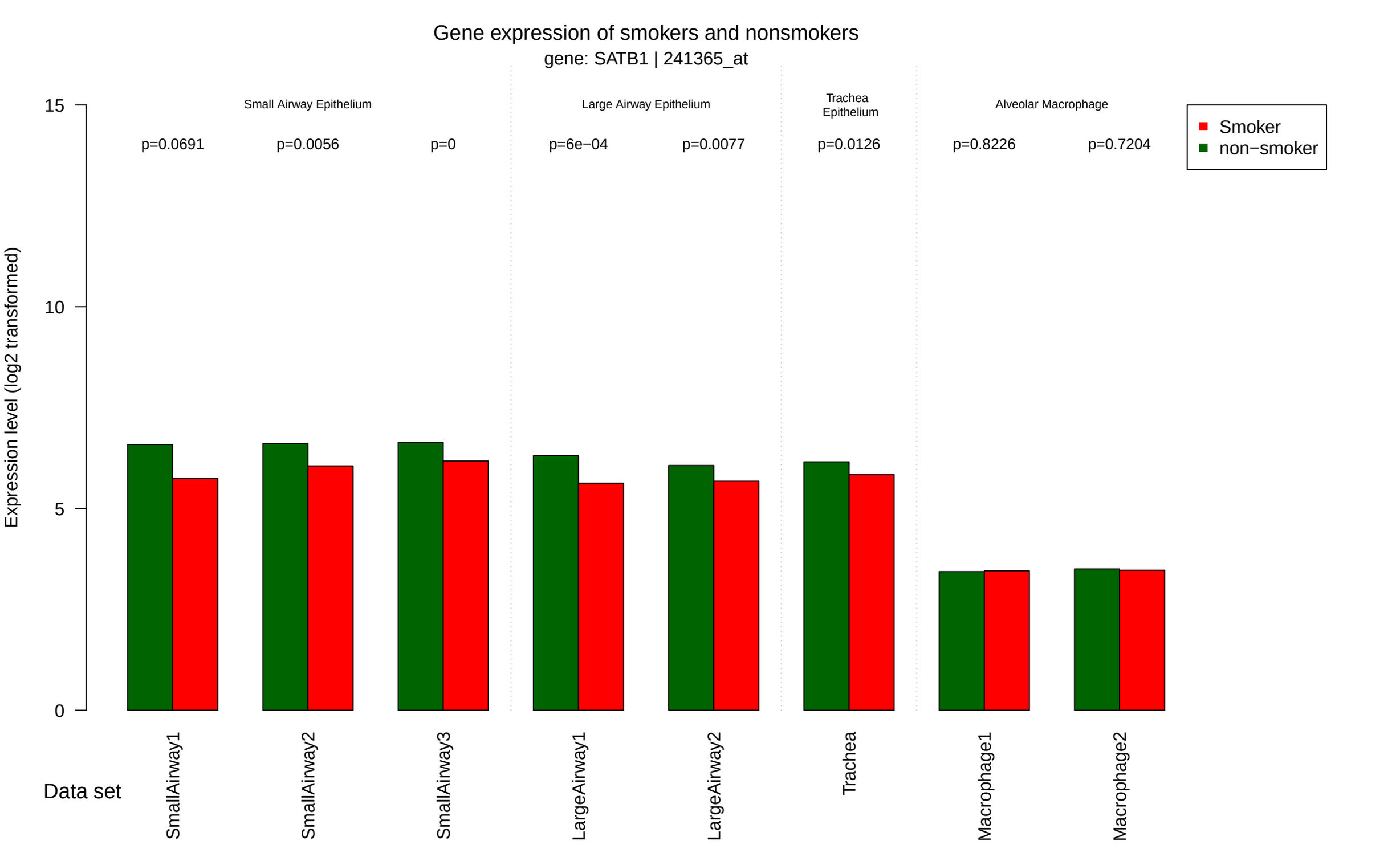
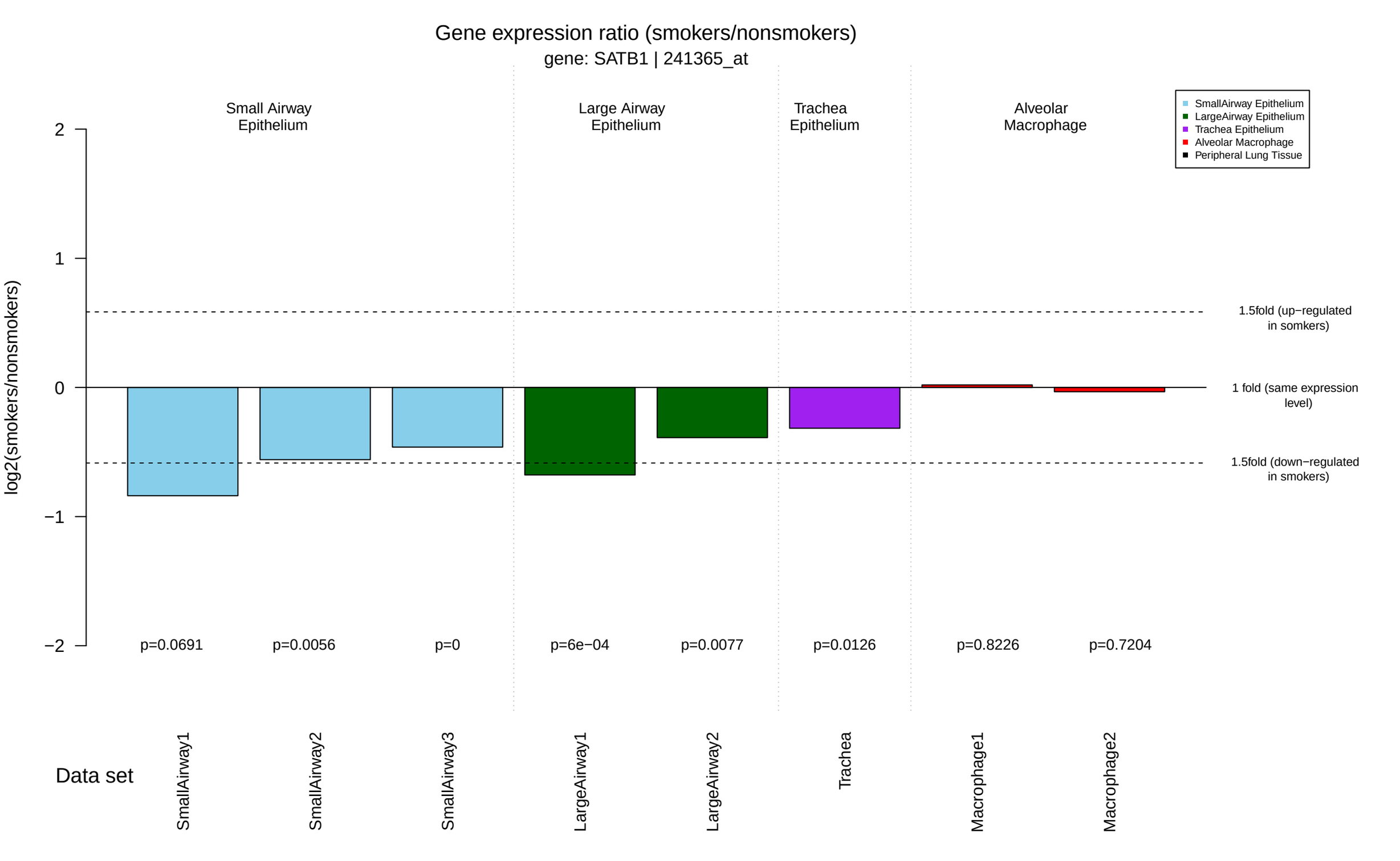
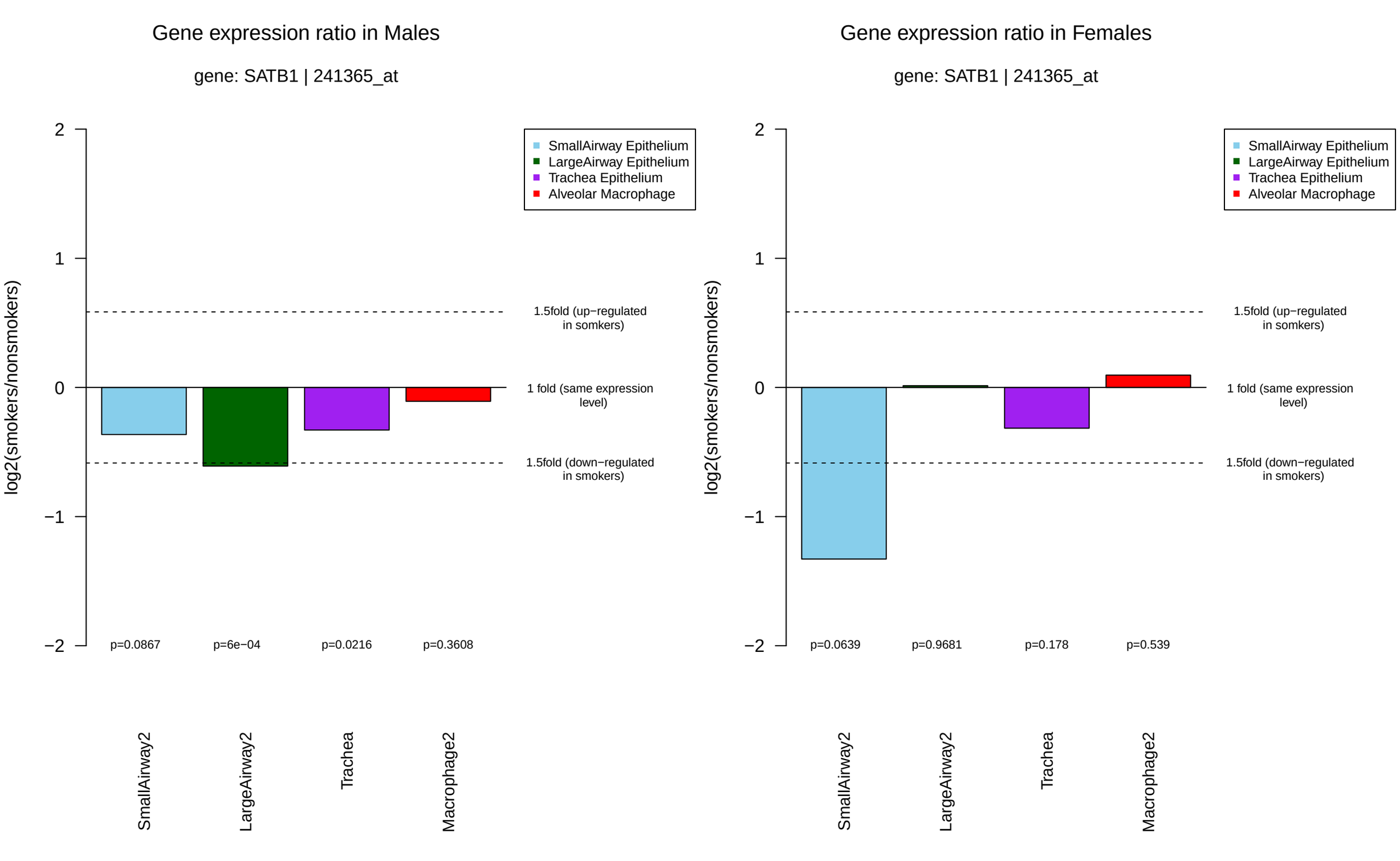
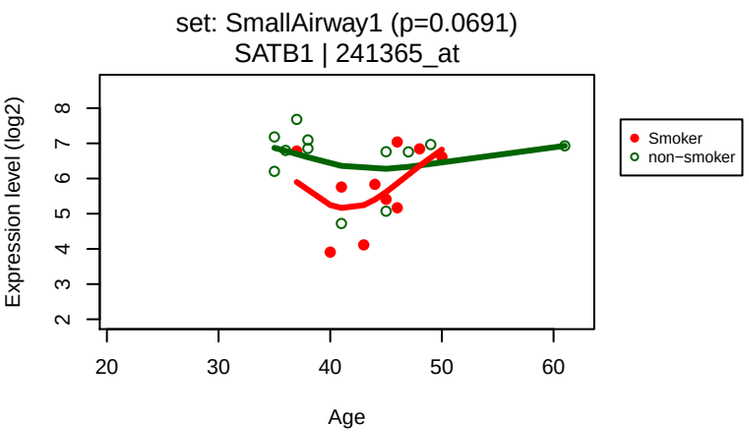
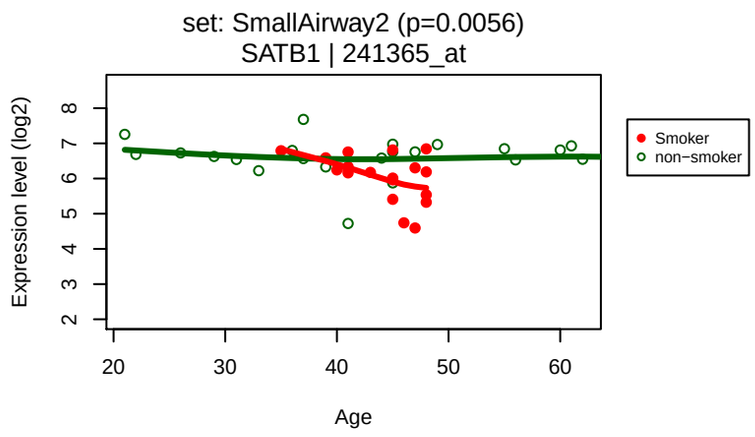
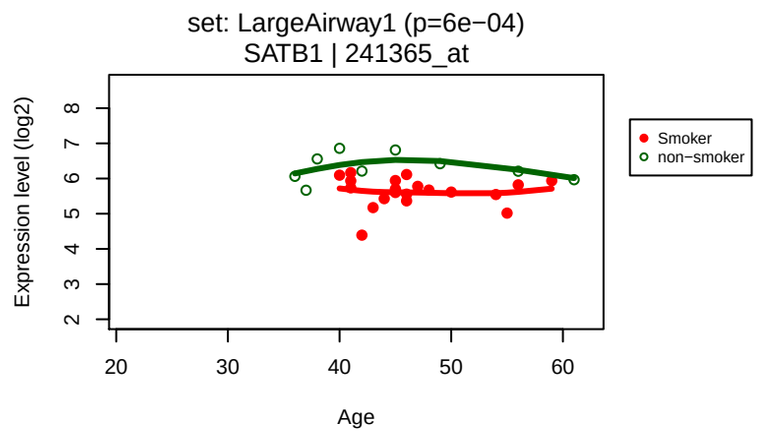

Gene expression of smokers and nonsmokers
gene: SATB1 | 241365_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0691
p=0.0056
p=0
p=6e−04
p=0.0077
p=0.0126
p=0.8226
p=0.7204
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

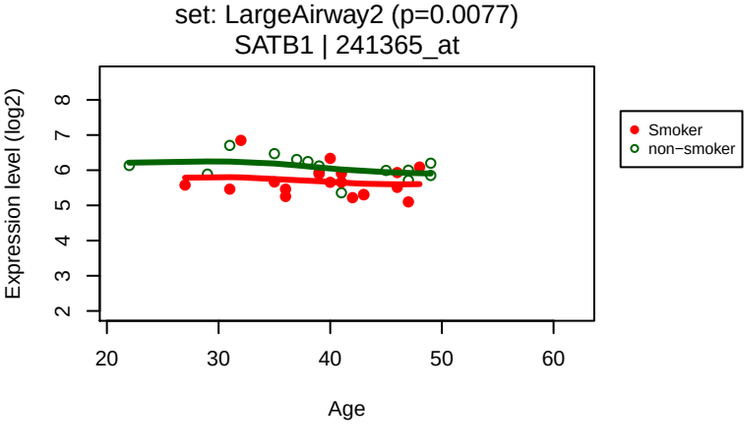

Gene expression ratio (smokers/nonsmokers)
gene: SATB1 | 241365_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0691
p=0.0056
p=0
p=6e−04
p=0.0077
p=0.0126
p=0.8226
p=0.7204
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

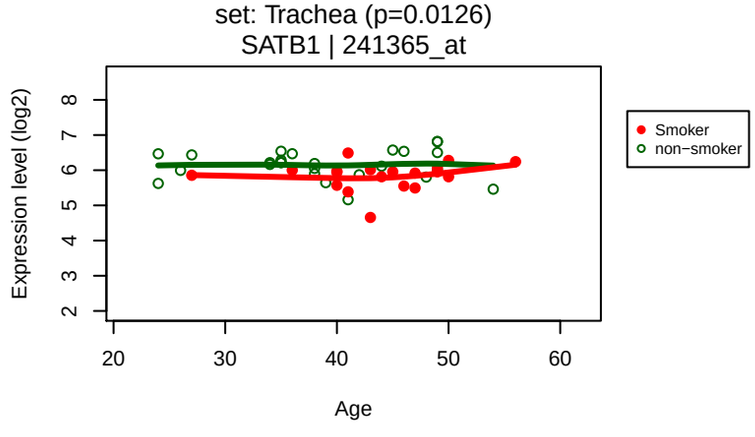

Gene expression ratio in Males
gene: SATB1 | 241365_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0867
p=6e−04
p=0.0216
p=0.3608
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SATB1 | 241365_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0639
p=0.9681
p=0.178
p=0.539
SmallAirway2
LargeAirway2
Trachea
Macrophage2

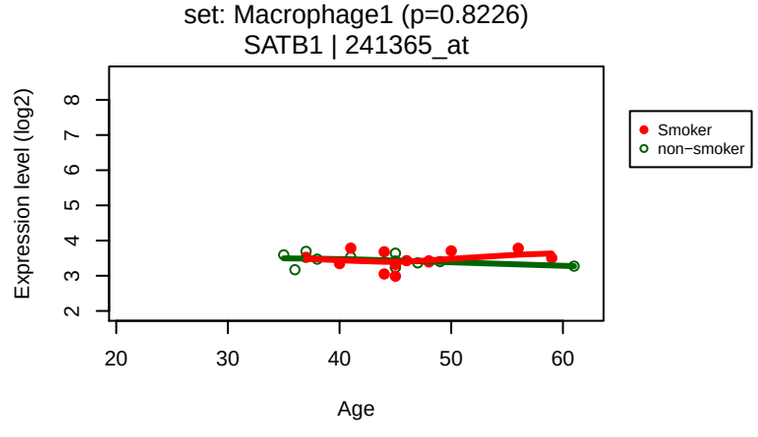

set: SmallAirway1 (p=0.0691)
SATB1 | 241365_at
Expression level (log2)
Age
Smoker
non−smoker

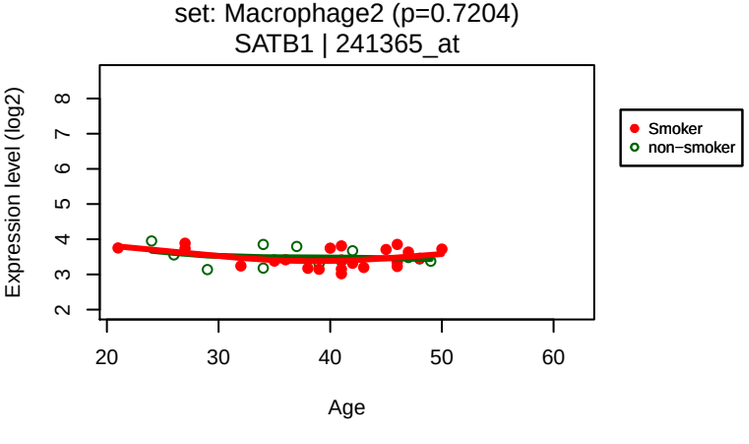

set: SmallAirway2 (p=0.0056)
SATB1 | 241365_at
Expression level (log2)
Age
Smoker
non−smoker

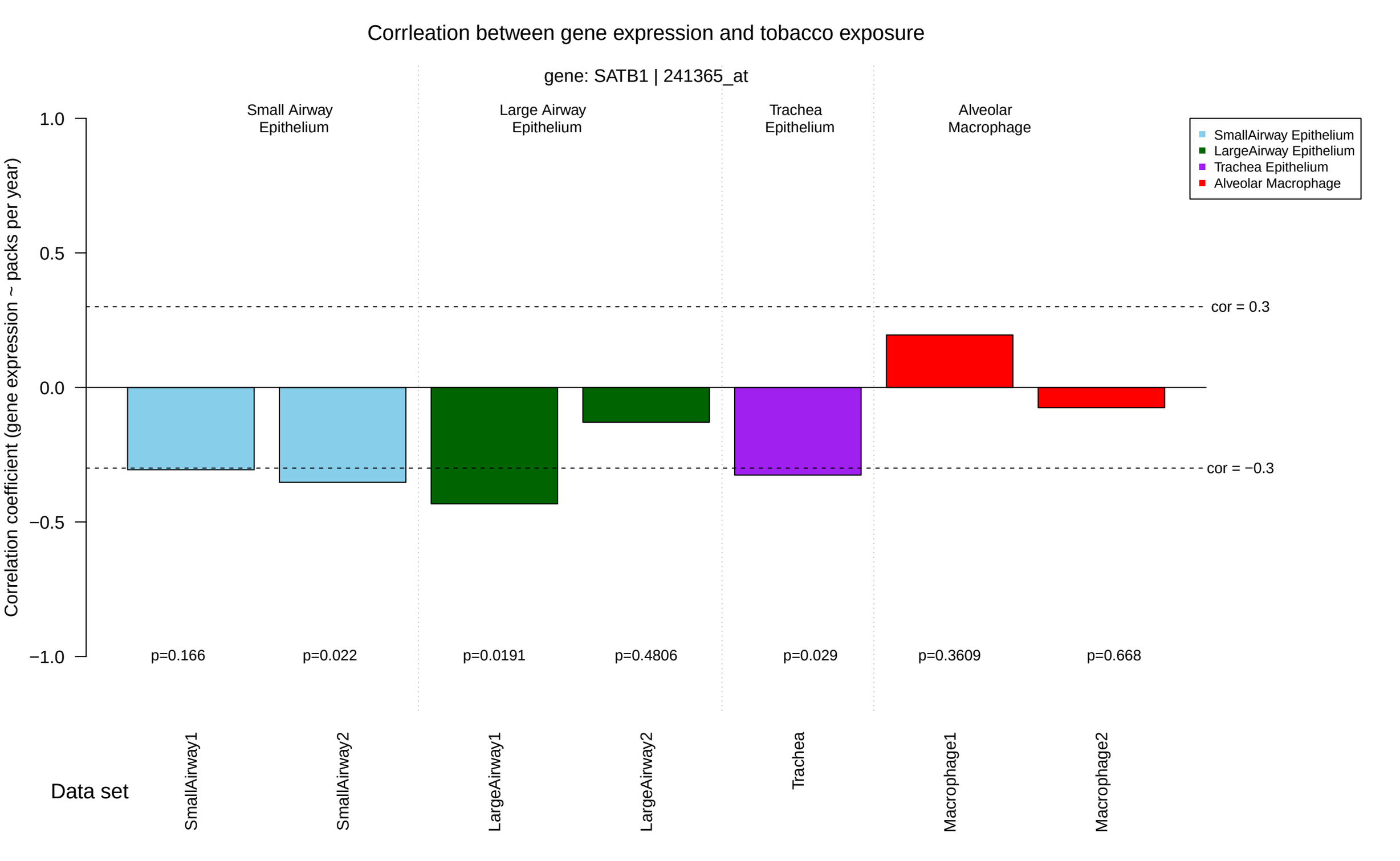

set: LargeAirway1 (p=6e−04)
SATB1 | 241365_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0077)
SATB1 | 241365_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0126)
SATB1 | 241365_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8226)
SATB1 | 241365_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7204)
SATB1 | 241365_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SATB1 | 241365_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.166
p=0.022
p=0.0191
p=0.4806
p=0.029
p=0.3609
p=0.668
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2